Segmentation and intervention sizing

Full method, segment structure, regional and age breakdowns, and the sizing behind each candidate intervention.

Provenance. The source file is the widely circulated Medical Cost Personal Datasets teaching set, which is simulated rather than drawn from real claims. The findings are real properties of the data and the methods are the ones that would be used on a live book, but no conclusion here describes an actual insurer or population.

Method

The source file contains 1,338 records and 7 columns with no missing values. One exact duplicate pair was found — two byte-identical rows describing a 19-year-old male non-smoker in the northwest — and one copy was removed, leaving 1,337 records. With no member identifier there is no way to distinguish a genuine coincidence from a duplication error; removing it is the conservative choice and it moves no published figure materially.

Members were split into four segments on two binary flags: smoking status as declared, and BMI at or above 30, the clinical obesity threshold. No other transformation was applied.

Segment structure

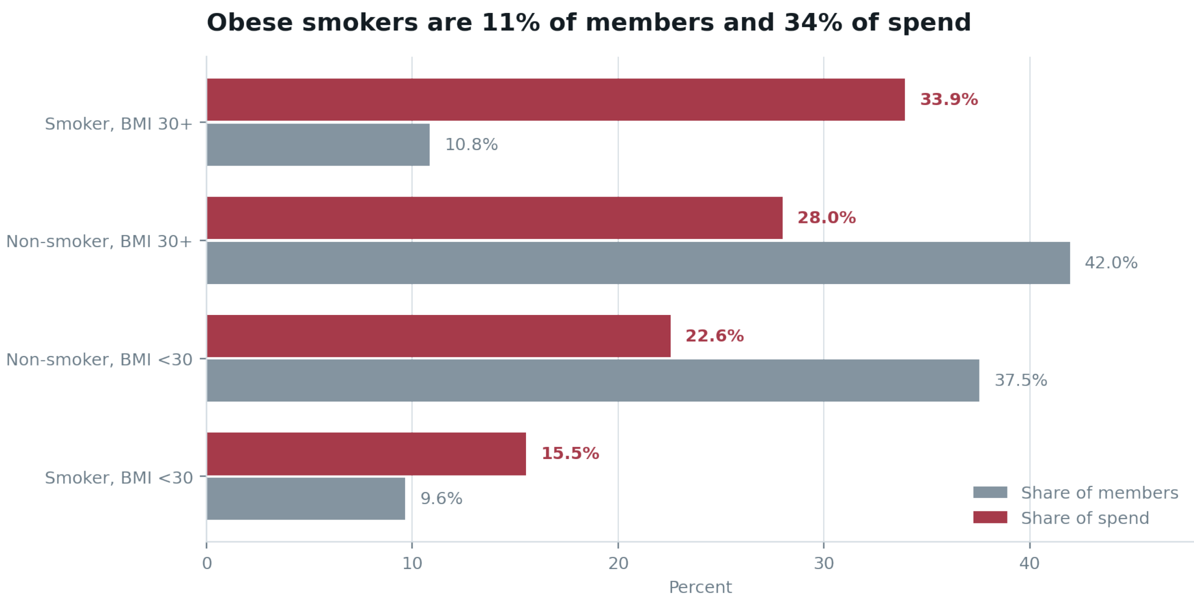
Segment	n	% mem	Mean	Median	Spend	% spend
Smoker, BMI 30+	145	10.8%	$41,558	$40,904	$6.03M	33.9%
Non-smoker, BMI 30+	561	42.0%	$8,856	$8,084	$4.97M	28.0%
Non-smoker, BMI <30	502	37.5%	$7,977	$6,762	$4.00M	22.6%
Smoker, BMI <30	129	9.6%	$21,363	$20,167	$2.76M	15.5%

The mean-to-median gap is the segment's internal skew. It is widest among obese smokers, where a small number of very large claims pull the mean above the typical member.

Spend is concentrated: the top 5% of members account for 17.3% of the book, the top 10% for 31.8%, and the top 20% for 51.6%. Charges are right-skewed with a skew coefficient of 1.52: median $9,386 against a mean of $13,279 and a maximum of $63,770. Any figure reported as a mean without its median will overstate the typical member.

The interaction

Moving from BMI under 30 to BMI 30 or above costs a non-smoker $879. The same move costs a smoker $20,195. The risk factors are not additive, and a targeting model that treats them as independent will mis-rank every segment.

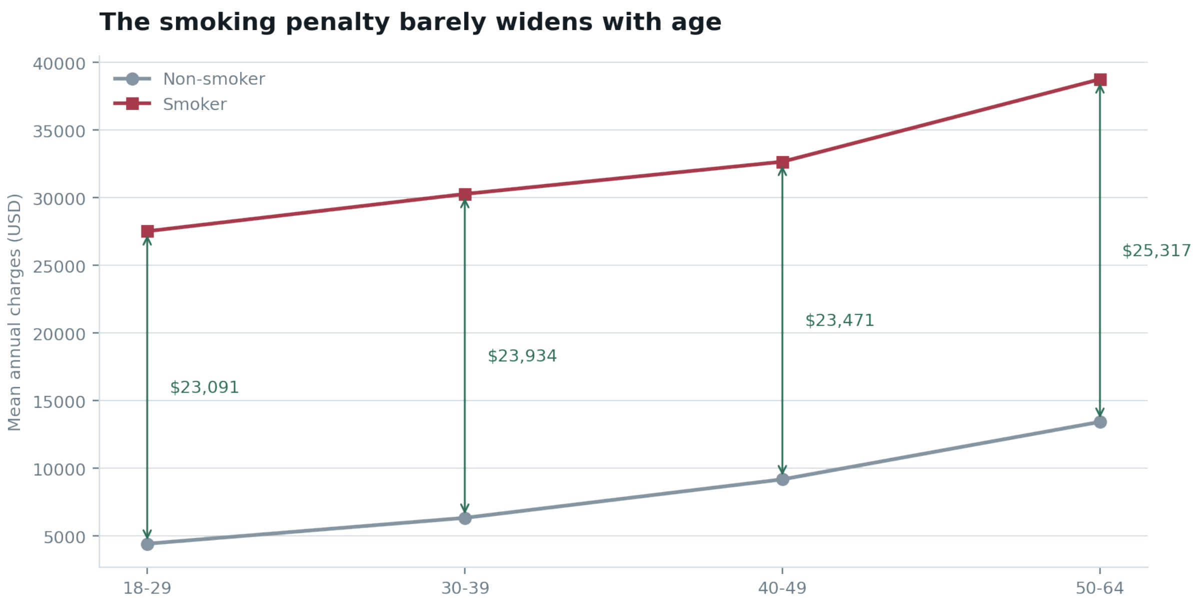


Population share against spend share, by segment.

Age is not the discriminator it appears to be

Both smoker and non-smoker costs rise with age, which invites the conclusion that older smokers are the priority. The gap between the two lines tells a different story.

Age band	n non-smoker	n smoker	Mean non-smoker	Mean smoker	Gap
18-29	330	86	$4,427	$27,518	$23,091
30-39	199	58	$6,337	$30,271	$23,934
40-49	217	62	$9,183	$32,655	$23,471
50-64	317	68	$13,431	$38,748	$25,317



The smoking penalty varies by roughly $2,200 across a 46-year age range. For practical purposes it is constant.

The implication runs against intuition. If a smoker costs about $24,000 more per year than a non-smoker regardless of age, then a successful cessation at 25 banks that difference for forty years and a cessation at 60 banks it for five. Targeting should skew young, not old.

Regional distribution

Region	n	Smoker %	Obese %	Mean BMI	Obese smoker %	Mean charges
northeast	324	20.7%	44.1%	29.2	9.0%	$13,406

Region	n	Smoker %	Obese %	Mean BMI	Obese smoker %	Mean charges
northwest	324	17.9%	45.4%	29.2	7.1%	$12,451
southeast	364	25.0%	66.8%	33.4	15.9%	$14,735
southwest	325	17.8%	53.2%	30.6	10.8%	$12,347

The southeast carries nearly double the obese-smoker prevalence of any other region, and that segment alone accounts for 45.7% of southeast spend against 28.9% elsewhere. Whether this reflects population composition or differences in care access cannot be settled from this file; it needs external deprivation or access data.

Intervention sizing

Each lever moves its segment onto the observed mean of the segment it would belong to with the risk factor removed. The ceiling is the resulting difference across the whole segment. Uptake columns apply a flat success rate to that ceiling.

Lever	n	Ceiling	Per head	At 5%	At 10%	At 20%
Smoking cessation — obese smokers	145	$4.74M	$32,702	$237k	$474k	$948k
Smoking cessation — non-obese smokers	129	$1.73M	$13,386	$86k	$173k	$345k
Weight programme — obese smokers	145	$2.93M	$20,195	$146k	$293k	$586k
Weight programme — obese non-smokers	561	$0.49M	$879	$25k	$49k	$99k

> **These are not causal effects.** The dataset has no time dimension, so nothing here can demonstrate that removing a risk factor causes a cost reduction. Every saving figure is an association-based ceiling: it moves a segment onto the observed mean of the segment it would belong to without that risk factor, assuming complete conversion and no residual risk.

Ranking

Levers are scored on impact multiplied by confidence and divided by effort. Confidence reflects how well the underlying effect is established in this data; effort reflects the operational difficulty of delivering a sustained change.

Rank	Lever	At 10%	Confidence	Effort	Reasoning
1	Cessation — obese smokers	$474k	High	Medium	Largest effect, smallest segment, established intervention.
2	Cessation — non-obese smokers	$173k	High	Medium	Same mechanism, roughly a third of the yield.
3	Instrument the data	—	High	Low	Cheap, and it unlocks the 25% of variance nothing here explains.

Rank	Lever	At 10%	Confidence	Effort	Reasoning
4	Weight — obese smokers	$293k	Low	High	Rests entirely on the BMI 30 artifact. Do not fund on this basis.
5	Weight — obese non-smokers	$49k	Medium	High	Largest population, smallest per-head return.

The ranking says target 11% of members to address 34% of spend, and it puts the largest population last. That is the whole finding.